

33 Chapter Review

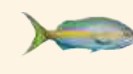


➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

This table recaps the animal groups surveyed in this chapter.

Key Concept	Phylum	Description
CONCEPT 33.1 Sponges are basal animals that lack tissues (pp. 690–691) ? Lacking tissues and organs, how do sponges accomplish tasks such as gas exchange, nutrient transport, and waste disposal?	Porifera (sponges)	 Lack tissues; have choanocytes (collar cells—flagellated cells that ingest bacteria and tiny food particles)
CONCEPT 33.2 Cnidarians are an ancient phylum of eumetazoans (pp. 691–693) ? Describe the cnidarian body plan and its two major variations.	Cnidaria (hydras, jellies, sea anemones, corals)	 Unique stinging structures (nematocysts) housed in specialized cells (cnidocytes); diploblastic; radially symmetrical; gastrovascular cavity (digestive compartment with a single opening)
CONCEPT 33.3 Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms (pp. 694–705) ? Is the lophotrochozoan clade united by unique morphological features shared by all of its members? Explain.	Metazoa { Eumetazoa { Lophotrochozoa { Platyhelminthes (flatworms) Syndermata (rotifers and acanthocephalans) Ectoprocta and Brachiopoda Mollusca (clams, snails, squids) Annelida (segmented worms)	 No body cavity; dorsoventrally flattened; gastrovascular cavity or no digestive tract  Hemocoel; rotifers have alimentary canal (digestive tube with mouth and anus) and jaws (trophi); acanthocephalans are parasites of vertebrates  Coelom; have lophophores (feeding structures bearing ciliated tentacles)  Hemocoel; reduced coelom; three main body parts (muscular foot, visceral mass, mantle); most have hard shell made of calcium carbonate  Coelom; body wall and internal organs are segmented (except digestive tract, which is unsegmented)
CONCEPT 33.4 Ecdysozoans are the most species-rich animal group (pp. 705–713) ? Describe some ecological roles of nematodes and arthropods.	Eumetazoa { Bilateria { Ecdysozoa { Nematoda (roundworms) Arthropoda (spiders, centipedes, crustaceans, and insects)	 Hemocoel; cylindrical body with tapered ends; no circulatory system; undergo ecdysis  Hemocoel; reduced coelom. Have segmented body, jointed appendages, and exoskeleton made of protein and chitin
CONCEPT 33.5 Echinoderms and chordates are deuterostomes (pp. 713–715) ? You've read that echinoderms and chordates are closely related and have evolved independently for over 500 million years. Explain how both of these statements can be correct.	Deuterostomia { Echinodermata (sea stars, sea urchins) Chordata (lancelets, tunicates, vertebrates)	 Coelom; bilaterally symmetrical larvae and five-part body organization as adults; unique water vascular system; endoskeleton  Coelom; have notochord; dorsal, hollow nerve cord; pharyngeal slits; post-anal tail (see Figure 34.3)